

Supplement of AOI-Net: Structural Face AOI-Guided Eye-Gaze Track Representation Learning for Autism Spectrum Disorder Detection

Abstract—This supplementary document provides supporting details and extended evidence for the proposed AOI-Net framework. The eight eye-movement datasets and their collection settings are introduced, including participant demographics, the eye-tracking configuration and preprocessing pipeline, and the ASD/TD sample composition of each paradigm. The complete training procedure of AOI-Net is then detailed in Algorithm 1, covering temporal encoding, AOI graph construction, structural encoding, dual-expert fusion, classification, loss computation, and the overall optimization scheme. Four sets of experiments are further reported: sensitivity analysis of the hyperparameter η , gradient-based analysis of AOI importance, an ablation study on fusion strategies and gating activations, and ROC analyses on all eight datasets. These results further demonstrate the robustness, interpretability, and discriminative ability of AOI-Net for ASD screening.

I. DATASET

This study utilizes eight eye-movement datasets, namely Speaking, Walking1, Walking2, Helicopter, Baby, Tablet, Attention, and Sad, collected by the Shenzhen Maternity and Child Healthcare Hospital. These datasets were designed to capture children's gaze behaviors under different dynamic video viewing scenarios, providing diverse eye-movement patterns for ASD-related analysis. Participants were diagnosed through comprehensive clinical evaluations, including standardized behavioral assessments and expert evaluations conducted by licensed clinicians. The TD group was screened to exclude known neurodevelopmental disorders. Detailed demographic and clinical characteristics of the participants are summarized in Table s-I.

During data collection, participants were seated approximately 57–65 cm from the display, and eye movements were recorded using an SMI eye-tracking system with a sampling rate of 250 Hz. A standard calibration procedure was performed before each recording session to ensure gaze measurement accuracy. All participants were presented with identical dynamic video stimuli under the same experimental protocol, with each video corresponding to a fixed-duration viewing task. The hardware configuration and key preprocessing parameters are summarized in Table s-II.

The raw gaze signals were converted into fixation sequences using a standard fixation detection procedure, where a fixation was defined as a sequence of consecutive gaze samples lasting at least 100 ms. During preprocessing, invalid gaze samples caused by blinks or tracking loss were removed, and gaze points outside the valid display region were filtered. The resulting fixation sequences were subsequently mapped to predefined areas of interest (AOIs), which were manually annotated on video frames and consistently applied across all participants. These AOI-based representations preserve the

TABLE s-I
PARTICIPANT DEMOGRAPHICS AND CLINICAL CHARACTERISTICS OF THE DATASET.

Characteristic	Value
Age of ASD group (years)	1.93 ± 0.73
Age of TD group (years)	1.74 ± 0.57
Gender distribution of ASD group	Male: 81.8%, Female: 18.2%
Gender distribution of TD group	Male: 54.1%, Female: 45.9%

TABLE s-II
EYE-TRACKING HARDWARE CONFIGURATION AND PREPROCESSING PARAMETERS.

Category	Description
Eye-tracking device	SMI eye tracker
Sampling rate	250 Hz
Eye tracking mode	Binocular tracking
Operating distance	57–65 cm
Stimulus type	Dynamic video stimuli
Fixation definition	Consecutive gaze samples exceeding 100 ms
Calibration protocol	Smart calibration procedure provided by the SMI eye-tracking system

TABLE s-III
DATASET COMPOSITION: THE TABLE SUMMARIZES THE NUMBER OF PARTICIPANTS IN THE DATASET, CATEGORIZED BY ASD AND TD INDIVIDUALS. IR DENOTES THE IMBALANCE RATIO.

Index	Dataset	Participants	ASD	TD	IR
1	Speaking	1390	966	424	2.28
2	Walking1	1356	939	417	2.25
3	Walking2	1380	957	423	2.26
4	Helicopter	1375	955	420	2.27
5	Baby	1348	939	409	2.30
6	Tablet	1372	954	418	2.28
7	Attention	1349	939	410	2.29
8	Sad	1277	882	395	2.23

semantic relationships among visual regions and provide the structural information used by the proposed framework.

The dataset is designed to facilitate the discrimination of children with ASD from typically developing (TD) children. As an initial step in the clinical assessment, ASD-versus-TD identification provides a foundation for subsequent evaluations, including subtype characterization, comorbidity analysis, and individualized assessment. The numbers of ASD and TD samples for each experimental paradigm, together with the corresponding imbalance ratios ($IR = N_{ASD}/N_{TD}$), are reported in Table s-III.

II. ALGORITHM

Algorithm 1 summaries the overall training procedure. Specifically, in **Temporal Encoding** (Line 4), the temporal

Algorithm 1: Training Procedure of AOI-Net.

Input: Fixation sequence $S = \{s_1, \dots, s_T\}$, AOI labels $A = \{a_1, \dots, a_T\}$

Output: Model parameters θ

- 1 Initialize parameters θ ;
- 2 **for** each training epoch **do**
- 3 **for** each mini-batch (S, A, y) **do**
- 4 **Temporal encoding:**
 $\mathbf{h} \leftarrow \text{TemporalEncoder}(S)$;
- 5 **AOI graph construction:**
Add temporal edges: $(v_t, v_{t+1}) \in \mathcal{E}_{\text{Temp}}$;
Add AOI edges: $(v_i, v_j) \in \mathcal{E}_{\text{AOI}}$ if $a_i = a_j$;
Construct edge set: $\mathcal{E} \leftarrow \mathcal{E}_{\text{Temp}} \cup \mathcal{E}_{\text{AOI}}$;
Form graph structure: $\mathcal{G} \leftarrow (\mathcal{V}, \mathcal{E})$;
- 6 **Structural encoding:** $\mathbf{g} \leftarrow \text{GNN}(\mathcal{G})$;
- 7 **Dual-expert fusion:**
 $(\omega_{TE}, \omega_{SE}) \leftarrow \text{Softmax}(\text{Gate}(S))$;
 $\mathbf{z} \leftarrow \omega_{TE} \cdot \mathbf{h} + \omega_{SE} \cdot \mathbf{g}$;
- 8 **Classification:** $\hat{y} \leftarrow \text{Classifier}(\mathbf{z})$;
- 9 **Loss computation:**
 $L_{Im} \leftarrow \eta L_{CaR} + (1 - \eta) L_{Ia}$;
 $L_{Total} \leftarrow L_{Im} + \mu L_{LB}$;
- 10 **Parameter update:** $\theta \leftarrow \text{Adam}(\theta, \nabla L_{Total})$;
- 11 **end**
- 12 **end**

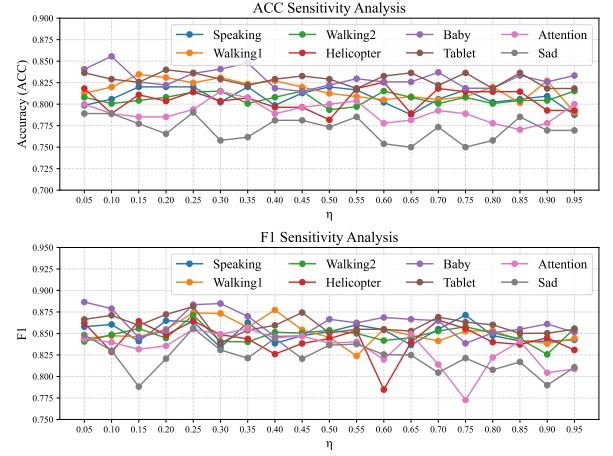


Fig. s-1. Sensitivity analysis of the hyperparameter η in terms of ACC and F1-score.

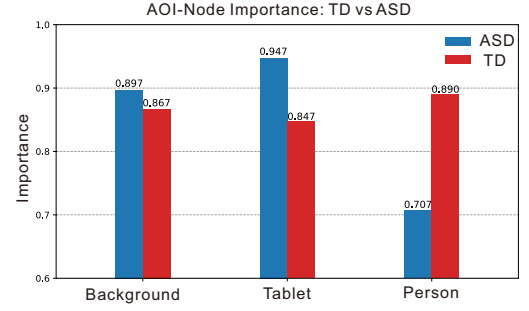


Fig. s-2. Average AOI node importance for the ASD and TD groups computed using a gradient-based interpretation method. The importance score reflects each AOI node's contribution to the final classification output, with higher values indicating greater influence on the model decision.

average performance and is therefore used in the subsequent experiments.

B. Analysis of AOI Importance and Prototype Behavior

To enhance the clinical interpretability of the proposed framework, we further analyze the contribution of each AOI node using a gradient-based attribution method, which quantifies how strongly the features of each AOI node contribute to the final decision. The result for the Tablet dataset is shown in Fig. s-2.

A distinct divergence in importance distribution can be observed between the two groups. For TD samples, the model predominantly assigns higher importance to the AOI corresponding to the “Person”, indicating a stronger reliance on socially relevant cues. In contrast, when discriminating ASD samples, the network places greater emphasis on non-social regions, particularly the “Background” and the “Tablet”. This tendency is consistent with well-documented atypical attentional preferences in ASD, where individuals are more likely to allocate gaze to objects or environmental details rather than socially salient information.

III. SUPPLEMENTARY EXPERIMENTS

A. Sensitivity Analysis of the Hyperparameter

To evaluate the robustness of AOI-Net with respect to the hyperparameter η , we conducted a sensitivity analysis by varying η while keeping all other hyperparameters fixed. As shown in Fig. s-1, both ACC and F1-score exhibit limited variation across a broad range of η values on all datasets, indicating that the model is relatively insensitive to this hyperparameter. The results also suggest that exhaustive hyperparameter tuning may not be necessary to obtain competitive performance. Among the evaluated settings, $\eta = 0.25$ achieves the best

TABLE s-IV

FUSION STRATEGIES AND GATE ACTIVATIONS IN THE MOE MODULE.
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Dataset	Metric	Fusion Strategy			Activation Function	
		Cross-attention	Concatenation	Mean Fusion	AOI-Net (Sigmoid)	AOI-Net (Softmax)
Speaking	ACC	0.8141	0.8034	0.8249	0.8225	0.8201
	F1	0.8698	0.8612	0.8797	<u>0.8769</u>	0.8637
Walking1	ACC	0.7953	0.8150	0.8162	0.8297	<u>0.8238</u>
	F1	0.8455	0.8646	0.8611	0.8775	<u>0.8739</u>
Walking2	ACC	0.8007	0.7971	0.8056	0.8068	0.8140
	F1	0.8615	0.8519	0.8639	<u>0.8680</u>	0.8714
Helicopter	ACC	0.7830	0.7964	0.8085	<u>0.8097</u>	0.8145
	F1	0.8444	0.8542	0.8617	0.8654	<u>0.8647</u>
Baby	ACC	0.8235	0.8111	<u>0.8321</u>	0.8185	0.8358
	F1	0.8706	0.8652	<u>0.8790</u>	0.8689	0.8835
Tablet	ACC	0.8182	0.8206	0.8327	<u>0.8339</u>	0.8364
	F1	0.8649	0.8666	0.8746	<u>0.8798</u>	0.8813
Attention	ACC	0.7580	0.7741	0.7790	<u>0.7877</u>	0.7938
	F1	0.8319	0.8407	<u>0.8494</u>	0.8483	0.8551
Sad	ACC	0.7461	0.7578	0.7721	<u>0.7760</u>	0.7904
	F1	0.8179	0.8225	0.8439	<u>0.8451</u>	0.8561
Average Rank	ACC	4.50	4.38	2.63	<u>2.13</u>	1.38
	F1	4.38	4.25	2.63	<u>2.13</u>	1.63

C. Analysis of Fusion Strategies and Gating Mechanisms

To validate the effectiveness of the proposed gated-sum MoE fusion strategy, we compared it with several representative fusion schemes, including concatenation, mean fusion, and cross-attention. We also replaced the softmax-based gating mechanism with sigmoid activation to investigate the influence of gate functions. The results are summarized in Table s-IV. AOI-Net achieves competitive performance with both softmax- and sigmoid-based gating mechanisms, with softmax gating generally providing more stable improvements across datasets, demonstrating the effectiveness of adaptive expert fusion for integrating temporal and AOI representations. Fixed fusion strategies, including concatenation and mean fusion, remain competitive in some cases but cannot adjust expert contributions according to different gaze patterns. Cross-attention-based fusion does not consistently outperform the lightweight gating mechanism, possibly because excessive feature interaction may weaken the specialization of temporal and structural experts. The sigmoid-based variant achieves comparable results, while softmax gating provides more stable improvements across tasks. This indicates that the main benefit comes from adaptive expert weighting rather than a specific activation function.

D. ROC Analysis on All Datasets

Due to space limitations, only the representative ROC curves on the Speaking and Walking1 datasets were reported in the main text. To further examine the screening capability of different methods across varying decision thresholds, we provide the ROC curves for all eight datasets in Figs. s-3–s-10. As shown in these figures, the ROC curves of AOI-Net generally lie above those of the competing approaches across most operating regions, indicating a superior and stable balance between Sensitivity and Specificity. Moreover, AOI-Net achieves the largest AUC values across all datasets,

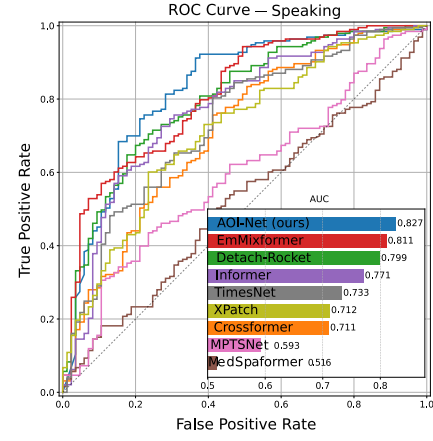


Fig. s-3. ROC curve and AUC of different methods on the Speaking dataset.

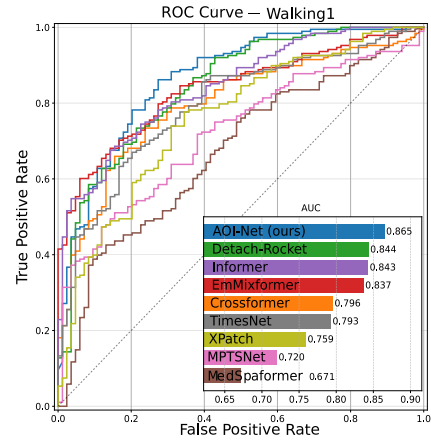


Fig. s-4. ROC curve and AUC of different methods on the Walking1 dataset.

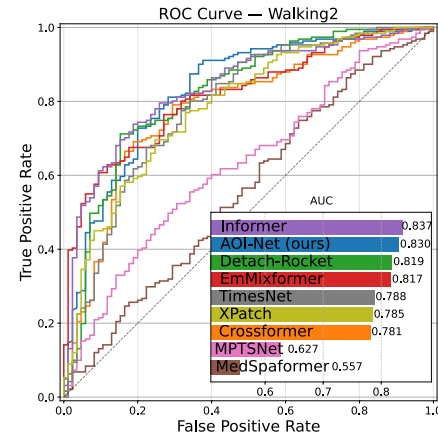


Fig. s-5. ROC curve and AUC of different methods on the Walking2 dataset.

demonstrating its more robust discriminative ability between ASD and TD under diverse experimental paradigms.

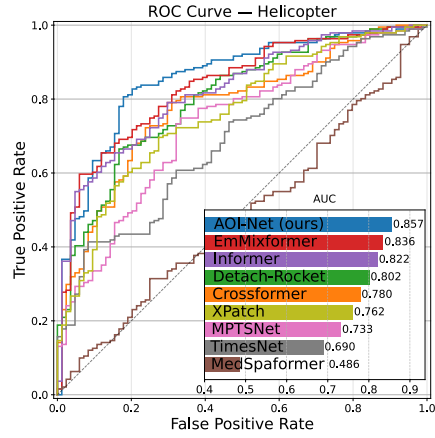


Fig. s-6. ROC curve and AUC of different methods on the Helicopter dataset.

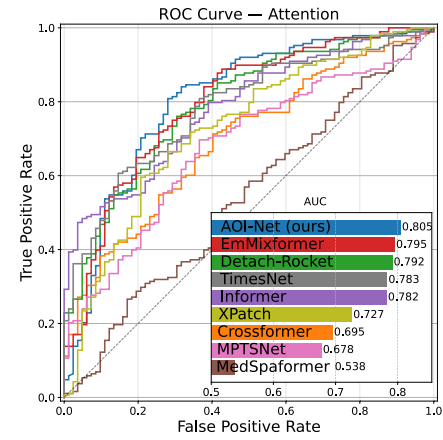


Fig. s-9. ROC curve and AUC of different methods on the Attention dataset.

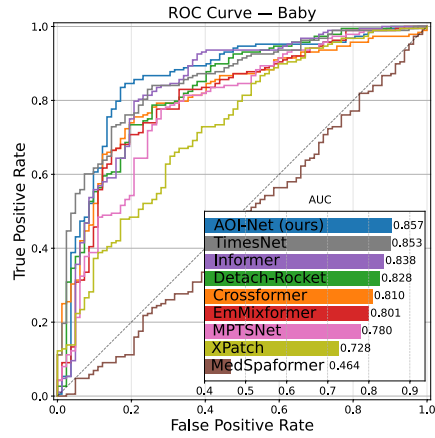


Fig. s-7. ROC curve and AUC of different methods on the Baby dataset.

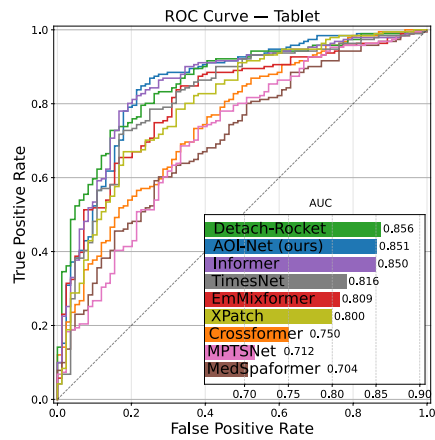


Fig. s-8. ROC curve and AUC of different methods on the Tablet dataset.

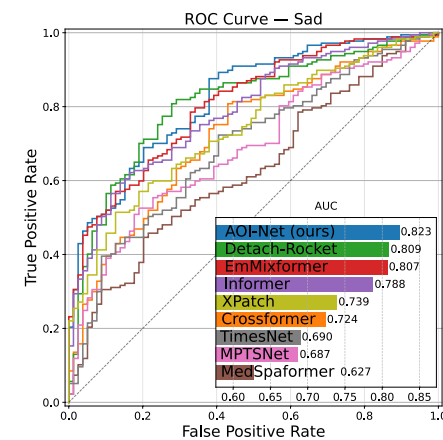


Fig. s-10. ROC curve and AUC of different methods on the Sad dataset.